Jatin Chowdhury

Audio Signal Processing Engineer

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Education

Massachusetts Institute of Technology *Cambridge, MA*

Ph.D in Electrical Engineering and Computer Science *September 2025 - PRESENT*

- Relevant research: real-time circuit simulation, machine learning, audio software development, signal processing.
- Relevant coursework: numerical algorithms, TinyML.

Stanford University, Center for Computer Research in Music and Acoustics *Palo Alto, CA*

M.A. in Music, Science, and Technology *September 2018 - June 2020*

- Relevant coursework: physical audio signal processing, time-frequency audio signal processing, machine learning, digital instrument design, spatial audio, music recording.
- Relevant research: physical modelling, nonlinear filter design, modal modelling, machine learning.
- Denning Family Fellowship for the 2018-2019 academic year.

University of Southern California *Los Angeles, CA*

B.S. in Electrical Engineering *August 2014 - May 2018*

- Completed Minors in Physics and Music Recording.
- Relevant coursework: signal processing, circuit design, digital logic, electromagnetics, software design.
- USC Presidential Scholarship for outstanding academic achievement.
- USC Renaissance Scholar certificate for excelling academically while pursuing separate fields of study.

Experience

Chowdhury DSP *Renton, WA*

Audio Signal Processing Engineer *October 2021 - PRESENT*

- Developed signal processing algorithms for real-time audio processing and synthesis.
- DSP algorithms include: virtual analog modelling, machine learning, filter design, and frequency-domain processing.
- Developed audio plugins including DSP, UI implementation, preset management, copy protection, installers, etc.
- Rendered services for clients including: BABY Audio, Schwabe Digital, Chandler Limited, Bogren Digital, Little Sir Rekko, VCV, KIT Plugins, Sabol Hill, Submission Audio, ADPTR Audio, and more.
- Hired, mentored, and trained interns in audio signal processing and software development.

Tesla Motors *Palo Alto, CA*

Audio Test Engineer *August 2020 - October 2021*

- Developed audio system signal flow layouts for vehicles using AudioWeaver.
- Contributed to audio system testing and tuning including telephony, equalization, and spatialization.
- Developed end-of-line tests to ensure vehicle audio system quality using Python and C++.
- Contributed to the signal processing and tuning for the vehicle pedestrian warning speaker.
- Contributed to testing and validation software for audio system firmware.

Persp3ctive VR *Los Angeles, CA*

Software/DSP Engineer *June 2019 - March 2020*

- Developed and implemented audio effects for use in a VR audio production environment.
- Implemented the audio effects as an audio plugin with a designed GUI.
- Built DSP framework for integrating effects made by other engineers into the plugin.
- Implemented OSC communication for sharing parameters and metering data with VR headset.

Audioworks Technologies *Toronto, ON*

Software Engineer *July 2018 - December 2018*

- Member of of the SoundsUnite developement team: building a digital audioworkstation (DAW) using JUCE/C++.
- Developed DSP features for the application including panning algorithms, level detection, and "smart" track exporting.
- Developed UX features for the application including meters, file browsers, and tempo controls.
- Integrating the application file management system with the SoundsUnite web store.

McGill Space Institute *Montreal, QC*
Research Trainee *May 2017 - August 2017*

- Member of the CHIME/FRB Working Group: building a software pipeline to detect Fast Radio Bursts (FRBs).
- Developed a "Flux Estimator" module for the pipeline to estimate the intrinsic brightness of astrophsyical events.
- Contributed to science and unit testing frameworks for the CHIME/FRB software pipeline.
- Presented work for the CHIME/FRB Working Group and CHIME/FRB Pipeline Gamma Release.

Skills

Programming Languages	C/C++, Python, Jai, Bash, MATLAB, Faust, LaTeX
Signal Processing	Virtual analog modelling, filter design, machine learning, time-frequency analysis
Frameworks	JUCE, CLAP, TensorFlow, PyTorch, Arduino
Tools	CMake, Git, Linux CLI, Visual Studio, Xcode, CI/CD, Inno Setup

Publications

Performance-Oriented Wave Digital Circuit Emulation *September 2026*
Proc. of the 29th International Conference on Digital Audio Effects *Cambridge, USA*

- Co-authored with Mark Rau.
- Available on the DAFx Archives.

Inference-Time Structured Pruning for Real-Time Neural Network Audio Effects *September 2025*
Proc. of the 28th International Conference on Digital Audio Effects *Ancona, IT*

- Co-authored with Christopher Johann Clarke.
- Available on the DAFx Archives.

Sample Rate Independent Recurrent Neural Networks for Audio Effects Processing *September 2024*
Proc. of the 27th International Conference on Digital Audio Effects *Guildford, UK*

- Co-authored with Alistair Carson, Alec, Wright, Vesa Välimäki, and Stefan Bilbao.
- Available on the DAFx Archives.

Computationally Efficient Physics Approximating Neural Networks for Highly Nonlinear Maps *October 2022*
Proc. of the Conference on Research in Adaptive and Convergent System *New York, USA*

- Co-authored with Christopher Johann Clarke and others.
- Available on the ACM Digital Library.

chowdsp_wdf: An Advanced C++ Library for Wave Digital Circuit Modelling *October 2022*
ArXiv E-Prints: Audio and Speech Processing *Online*

- Presents a wave digital circuit modelling library with fixed-topology optimizations and other advancements.
- Available on the ArXiv.

Emulating Diode Circuits with Differentiable Wave Digital Filters *June 2022*
19th Sound and Music Computing Conference *Saint Etienne, France*

- Co-authored with Christopher Johann Clarke.
- Available on Zenodo.

A Wave Digital Filter Modeling Library for the Faust Programming Language *June 2021*
Proc. of the 18th Sound and Music Computing Conference *Torino, IT*

- Co-authored with Dirk Roosenburg, Eli Stine, and Romain Michon.
- Available on GitHub.

RTNeural: Fast Neural Inferencing for Real-Time Systems *June 2021*
ArXiv E-Prints: Audio and Speech Processing *Online*

- Presents a neural network inferencing library for real-time systems.
- Available on the ArXiv.

Water Bottle Synthesis with Modal Signal Processing *September 2020*
Proc. of the 23rd International Conference on Digital Audio Effects *Vienna, Austria*

- Presented at the DAFx-2020 conference.
- Available on the DAFx Archives.

Stable Structures for Nonlinear Biquad Filters *September 2020*
Proc. of the 23rd International Conference on Digital Audio Effects *Vienna, Austria*
- Presented at the DAFx-2020 conference.
- Available on the DAFx Archives.

Real-Time Physical Modelling for Analog Tape Machines *September 2019*
Proc. of the 22nd International Conference on Digital Audio Effects *Birmingham, UK*
- Presented at the DAFx-2019 conference.
- Available on the DAFx Archives.

The CHIME Fast Radio Burst Project: System Overview *August 2018*
The Astrophysical Journal *Vol. 836, No. 1*
- Co-authored with the CHIME/FRB Collaboration
- Available on the ArXiv.

Professional & Academic Service

Peer Reviewer Provided peer review for papers submitted to conferences and journals, including the International Conference on Digital Audio Effects (DAFx), Audio Mostly, and the EURASIP Journal on Audio, Speech, and Music Processing.

Organizer Served on the local organizing committee for the 29th International Conference on Digital Audio Effects (DAFx26), in Cambridge, MA.

Mentor Participated in the Audio Developer Conference Mentorship Program in 2025 and 2026.

Presentations

Wait-Free Programming From Scratch *June 2026*
Northwest C++ User Group Meetup *Bellevue, WA*
- Discussed the challenges of designing wait-free algorithms for audio and other real-time software applications.
- Discussed the trade-offs of well-known solutions including spin mutexes and lock-free queues.
- Presented the development of a "from scratch" wait-free solution to a specific problem.

AI Inference for Real-Time Audio Applications *February 2026*
Boston AI Music Meetup *Cambridge, MA*
- Discussed the challenges of developing real-time systems in the context of music software.
- Discussed the challenges of performing neural network inference within a real-time system.
- Presented RTNeural as an inference engine capable of performing neural network inference in real-time.

Performance Engineering for Modern DSP *September 2025*
Signal Processing Summit *Milpitas, CA*
- Discussed performance engineering for signal processing engineers and software developers.
- Presented recommendations based on modern advancements in CPU and DSP hardware technology.
- Discussed ideas for teaching novice engineers about performance.

Real-Time Audio Programming with Memory Arenas *October 2024*
The Audio Programmer Monthly Meetup *Online*
- Presented a brief overview of heap and stack memory allocation.
- Introduced the concept of the "arena allocator" and some of its properties.
- Discussed the use of arena allocators for real-time audio programming applications.

Productive Programming with the JAI Programming Language *October 2024*
Northwest C++ User Group Meetup *Redmond, WA*
- Presented a brief introduction to the JAI programming language.
- Discussed relevant JAI features, including compile-time meta-programming and the JAI `Context`.
- Discussed the use of JAI in audio programming applications.

Real-Time Circuit Simulation with Wave Digital Filters In C++ (Poster) *September 2024*
CppCon *Aurora, CO*
- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ compile-time meta-programming for implementing WDFs.
- Compared the runtime performance of various WDFs written in C++ and other languages.
- Presented a simple synthesizer made entirely with WDFs.

A Comparison of Virtual Analog Modelling Techniques (Part 2) *November 2023*
Audio Developer Conference *London, UK*

- Co-presented with Christopher Johann Clarke.
- Presented a virtual analog model of the Tube Screamer distortion circuit using multiple modelling techniques.
- Discussed white-box methods: Modified Nodal Analysis, Nodal DK, and Wave Digital Filters with R-Type adaptors.
- Discussed black-box methods: Recurrent Neural Networks, conditional models, and transformers.
- Discussed grey-box methods: Symbolic Regression, Differentiable WDFs, and Physics-Approximating Neural Nets.

Audio Buffers, Buffer Views, and Buffer Iterators *September 2023*
The Audio Programmer Monthly Meetup *Online*

- Reviewed data structure design in C++, focusing on owning and non-owning containers.
- Discussed the design of an "owning" audio buffer data structure, and a "non-owning" buffer view data structure.
- Discussed the implementation of custom C++ iterators for audio buffer data structures.

Real-Time Circuit Simulation with Wave Digital Filters *July 2023*
Northwest C++ Users Group Meetup *Redmond, WA*

- Discussed the challenges of real-time audio programming.
- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ run-time polymorphism and compile-time meta-programming for implementing WDFs.
- Demonstrated example applications of WDF models in developing digital audio effects.

Replacing the AudioProcessorValueTreeState *December 2022*
The Audio Programmer Monthly Meetup *Online*

- Discussed the historical and current methods of audio plugin state management with the JUCE framework.
- Proposed a new approach for managing plugin state with an emphasis on real-time safety.

Physical Modelling for Analog Tape Emulation *December 2022*
Acoustical Society of America Annual Meeting *Nashville, USA*

- Discussed the development of a physical-modelling algorithm for real-time tape emulation.
- Discussed the physical basis for modelling magnetic hysteresis with nonlinear differential equations.
- Discussed implementation challenges for real-time tape emulation algorithms.

Analog Modelling with Wave Digital Filters in C++ *December 2021*
The Audio Programmer Monthly Meetup *London, UK*

- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Discussed two approaches for implementing WDFs in C++, with trade-offs between flexibility and performance.
- Demonstrated example applications of WDF models in developing digital audio effects.

CI/CD for Audio Plugin Development *November 2021*
Audio Developer Conference *London, UK*

- Discussed the basic concepts and motivations for continuous integration and continuous deployment (CI/CD).
- Discussed the usefulness of CI/CD for developing audio plugins.
- Demonstrated several CI/CD workflows in the context of audio plugin development.

Creating Audio Effect Plugins for the 21st Century *February 2021*
SARC Virtual Events Series *Online*

- Presented at the Sonic Arts Research Centre (SARC) at Queen's University Belfast.
- Presented a paradigm for developing original and innovative digital audio software.
- Discussed techniques and philosophies for designing audio software as part of a creative process, including signal processing algorithms and user interfaces.

A Comparison of Virtual Analog Modelling Techniques *November 2020*
Audio Developer Conference *London, UK*

- Presented a virtual analog model of the Klon Centaur guitar pedal developed using multiple modelling techniques, including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.
- An accompanying paper is published on the ArXiv.

Complex Nonlinearities for Audio Signal Processing *May 2019*
CCRMA DSP Seminar *Stanford, CA*

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented a series of audio effects constructed from innovative uses of nonlinear signal processing techniques.
- Audio effects presented include harmonic exciters, hysteresis, nonlinear filters, subharmonics generators, and more.

Exploring Real-Time DSP Systems for Mixing and Performance *April 2019*
CCRMA DSP Seminar *Stanford, CA*

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented overview of previous research and projects to an audience of peers, professors, and the public.
- Discussed topics including real-time timbral conversion, methods of integrating "imperfection" into audio DSP systems, and designing user interfaces for DSP systems as an extension of the DSP systems themselves.

Research

Code Generation Workflows for Circuit Emulation
Cambridge, MA

Lead Researcher
June 2025 - PRESENT

- Developed code generation systems for generating optimized implementations of circuit models using either Wave Digital Filters or nodal analysis.
- Developed a textual representation of Wave Digital circuit topology.
- Developed analysis tools for "reducing" circuit model complexity.

Real-Time Neural Network Inferencing
Stanford, CA

Lead Researcher
June 2020 - PRESENT

- Developed RTNeural, a C++ library for performing neural inferencing for real-time audio systems.
- Developed performance benchmarks to compare the library with LibTorch, ONNX Runtime, and TensorFlow Lite.
- Project documentation is available on GitHub.

Differentiable Wave Digital Filters

Lead Researcher
July 2021 - September 2022

- Demonstrated a process for constructing Wave Digital Filter (WDF) circuit models, in conjunction with neural network models of specific circuit elements.
- Developed a Python library for creating differentiable WDF models, allowing them to be trained via backpropagation.
- Project documentation is available on GitHub.

Klon Centaur Emulation
Stanford, CA

Lead Researcher
March 2020 - August 2020

- Developed a digital emulation of the Klon Centaur guitar distortion pedal.
- Used circuit modelling methods including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.
- Developed emulation as an audio plugin and guitar pedal using an Arduino Teensy microcontroller.
- Project documentation is available on GitHub.

Modal Waterbottles
Stanford, CA

Researcher
June 2019 - April 2020

- Worked with a team of researchers to develop acoustical models of water bottles using modal signal processing.
- Developed a real-time water bottle synthesizer using the JUCE framework.
- Project documentation is available on GitHub.

Complex Nonlinearities

Lead Researcher
August 2019 - June 2020

- Built frameworks for developing complex nonlinear audio signal processors.
- Developed a series of open-source plugins to demonstrate the nonlinear frameworks.
- Published a series of Medium articles explaining nonlinear signal processing techniques for a non-technical audience.
- Project documentation can be found on GitHub.

Bad Circuit Modelling

Lead Researcher
November 2019 - June 2020

- Researched methods for virtual analog circuit modelling to account for imperfections that exist in real-world circuits.
- Developed simulations and real-time circuit-modelling systems that accurately model the aging and component tolerances of resistors and capacitors.
- Project documentation can be found on GitHub.

Analog Tape Modelling
Stanford, CA

Lead Researcher
January 2019 - April 2019

- Researched and developed a physical model of an analog, reel-to-reel tape machine.
- Modelled the nonlinear process of analog tape recording using the Jiles-Atherton model for magnetic hysteresis.
- Modelled other effects created by analog tape machines, including flutter, playhead loss effects, and biasing effects.
- Developed an open-source audio plugin implementing the analog tape model as a real-time system. Source code and documentation are available on GitHub.

notGuitar
Los Angeles, CA

Research Partner
January 2018 - May 2018

- notGuitar is real-time timbral conversion system designed to process a guitar input signal to sound like a saxophone.
- notGuitar was implemented using a Texas Instruments DSK6713 DSP board in May 2018.
- Documentation for the project can be found on GitHub.

Projects

Chowdhury DSP
Founder, Engineer *June 2018 - PRESENT*

- Developed open-source audio plugins including:
  - ChowTapeModel: An analog tape emulation based on the DAFx-19 paper "Real-Time Physical Modelling for Analog Tape Machines".
  - BYOD: A modular system for creating guitar effect signal chains.
  - ChowMatrix: A dynamic matrix of stereo delay effects.
  - ChowKick: A kick drum synthesizer based on creative circuit modelling.
  - ChowCentaur: A digital emulation of the Klon Centaur distortion pedal, using Wave Digital Filters and recurrent neural networks.
  - ChowPhaser: A phaser effect based loosely on the Schulte Compact Phasing 'A' circuit.
- Developed effects for guitar pedal platforms including the Aviate Audio Multiverse and Chaos Audio Stratus pedals.
- Contributed audio effects and other DSP code to the open-source Surge Synthesizer project.
- Developed modules for the VCV Rack modular synthesis environment.
- For more information, see the Chowdhury DSP website.

CLever Audio Plugins (CLAP)
Contributor *March 2022 - PRESENT*

- Contributed to the CLAP plugin standard with testing and documentation.
- Proposed new "extensions" to the CLAP standard.
- Contributed to a wrapper to allow developers using the JUCE framework to export CLAP plugins.
- Contributed to a JUCE module allowing JUCE applications to host CLAP plugins.

Wave Digital Filters
Programmer *January 2020 - PRESENT*

- Created a minimal C/C++ Library for modelling circuits with Wave Digital Filters (WDFs).
- Currently includes WDF implementations for basic circuit elements (resistors, capacitors, inductors), basic circuit adaptors (series, parallel), "1-port" nonlinear circuit elements, and "R-Type" adaptors.
- Includes several example WDF models implemented as audio plugins using the JUCE framework, as well as a graphical WDF prototyping tool.
- Developed an auxiliary script for deriving scattering matrices for "R-Type" adaptors.
- Ongoing development and documentation can be found on GitHub.

NoLava Recording Studios *Los Angeles, CA*
Audio Engineer *August 2017 - July 2018*

- Recorded, mixed, and mastered for artists of various styles including acoustic, electronic, rock, country, and punk.
- Installed, repaired, and maintained speakers, microphones, keyboards, amplifiers, and other studio equipment.